Day 047 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Use the Order of Operations

Standards: 6.EE.A.1, 6.EE.A.2c

Learning target: I can evaluate expressions using the order of operations.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Work inside parentheses, then evaluate exponents. Multiply and divide from left to right; then add and subtract from left to right. Multiplication does not automatically come before division.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 14, 18

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01 / Learn

Work inside parentheses, then evaluate exponents. Multiply and divide from left to right; then add and subtract from left to right. Multiplication does not automatically come before division.

Worked example 1

Evaluate 7 + 3 × 5. Show the intermediate steps.

1. Work inside parentheses, then evaluate exponents.

2. Multiply and divide from left to right; then add and subtract from left to right.

3. Use parentheses and exponents first, then × and ÷ left to right, then + and -.

4. Result: 22.

5. State the result in the form requested: 22.

Answer: 22

Worked example 2

Evaluate $5^2 + 3(8 - 3)$. Part A: enter the value. Part B: a student begins by adding 5^2 and 3; explain why that changes the expression.

1. Work inside parentheses, then evaluate exponents.

2. Parentheses: 8 - 3 = 5.

3. Exponent: $5^2 = 25$.

4. Multiply: 3 × 5 = 15.

5. Add: 25 + 15 = 40.

6. The product must be completed before the final sum.

Answer: 40

Day 047 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Evaluate 3 + 2 × 6. Show the intermediate steps.

02 / Constructed response

Evaluate 3 + 5 × 6. Show the intermediate steps.

03 / Constructed response

Evaluate 5 + 7 × 5. Show the intermediate steps.

04 / Constructed response

Evaluate (6 + 7) × 6^2. Show the intermediate steps.

05 / Constructed response

Evaluate (6 + 7) × 3^2. Show the intermediate steps.

06 / Constructed response

Evaluate (3 + 3) × 4^2. Show the intermediate steps.

07 / Constructed response

Evaluate 6 ÷ 3 × 6 + 3^2. Show the intermediate steps.

08 / Numeric entry and error analysis

Evaluate $2^2 + 4(7 - 3)$. Part A: enter the value. Part B: a student begins by adding 2^2 and 4; explain why that changes the expression.

09 / Numeric entry and error analysis

Evaluate $6^2 + 5(5 - 3)$. Part A: enter the value. Part B: a student begins by adding 6^2 and 5; explain why that changes the expression.

10 / Constructed response

Evaluate 30 ÷ 6 × 6 + 6^2. Show the intermediate steps.

Day 047 | Unit 04 | Answer key

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1. 15

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 15.

6.EE.A.1, 6.EE.A.2c | Constructed response

2. 33

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 33.

6.EE.A.1, 6.EE.A.2c | Constructed response

3. 40

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 40.

6.EE.A.1, 6.EE.A.2c | Constructed response

4. 468

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 468.

6.EE.A.1, 6.EE.A.2c | Constructed response

5. 117

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 117.

6.EE.A.1, 6.EE.A.2c | Constructed response

6. 96

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 96.

6.EE.A.1, 6.EE.A.2c | Constructed response

7. 21

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 21.

6.EE.A.1, 6.EE.A.2c | Constructed response

8. 20

Parentheses: 7 - 3 = 4. Exponent: $2^2 = 4$. Multiply: 4 × 4 = 16. Add: 4 + 16 = 20. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

9. 46

Parentheses: 5 - 3 = 2. Exponent: $6^2 = 36$. Multiply: 5 × 2 = 10. Add: 36 + 10 = 46. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

10. 66

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 66.

6.EE.A.1, 6.EE.A.2c | Constructed response